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United States Patent	12408368
Kind Code	B2
Date of Patent	September 02, 2025
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Method of making a semiconductor device using a dummy gate

Abstract

A method of making a semiconductor device includes forming a fin mask layer on a semiconductor layer, forming a dummy gate over the fin mask layer, and forming source and drain regions on opposite sides of the dummy gate. The dummy gate is removed and the underlying fin mask layer is used to define a plurality of fins in the semiconductor layer. A gate is formed over the plurality of fins.

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Appl. No.: 18/068718

Filed: December 20, 2022

Prior Publication Data

Document Identifier	Publication Date
US 20230121119 A1	Apr. 20, 2023

Related U.S. Application Data

continuation parent-doc US 15979326 20180514 ABANDONED child-doc US 18068718
continuation parent-doc US 15331714 20161021 US 9991351 20180605 child-doc US 15979326
continuation parent-doc US 14976781 20151221 US 9905662 20180227 child-doc US 15331714
division parent-doc US 13906789 20130531 ABANDONED child-doc US 14976781

Publication Classification

Int. Cl.: H10D30/62 (20250101); H10D30/01 (20250101); H10D30/69 (20250101); H10D62/13 (20250101); H10D62/822 (20250101); H10D64/01 (20250101); H10D64/66 (20250101); H10D84/83 (20250101)

U.S. Cl.:

CPC H10D30/6219 (20250101); H10D30/024 (20250101); H10D30/62 (20250101); H10D30/6211 (20250101); H10D30/797 (20250101); H10D62/151 (20250101); H10D62/822 (20250101); H10D64/017 (20250101); H10D64/018 (20250101); H10D64/661 (20250101); H10D84/834 (20250101);

Field of Classification Search

CPC: H10D (30/6219); H10D (30/6211); H10D (30/62); H10D (64/661); H10D (64/018); H10D (64/017)

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Primary Examiner: Ha; Nathan W

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This is a continuation of application Ser. No. 15/979,326, filed on May 14, 2018, which was a continuation of application Ser. No. 15/331,714, filed on Oct. 21, 2016 and issued as U.S. Pat. No. 9,991,351, which was a continuation of application Ser. No. 14/976,781, filed on Dec. 21, 2015 and issued as U.S. Pat. No. 9,905,662, which was a continuation of application Ser. No. 13/906,789, filed on May 31, 2013, the teachings of all of which are incorporated herein by reference in their entirety.

FIELD OF THE INVENTION

(1) The present invention relates to a method of making electronic devices, and more particularly, to a method of making semiconductor devices.

BACKGROUND

(2) Semiconductor device technologies continue to evolve, providing higher chip density and operating frequencies. Fin-type field-effect transistors (FinFETs) are one type of transistor technology that is currently used to help provide desired device scaling while maintaining appropriate power consumption budgets.

(3) A fin-type field effect transistor is a transistor that is formed with a fin of material. A fin is a relatively narrow width and relatively tall height structure that protrudes from the top surface of a semiconductor layer. The fin width is intentionally kept small to limit the short channel effect.

(4) In a conventional FinFET, a gate conductor is positioned on the top surface of the semiconductor layer and over a portion of the fin. The gate conductor runs parallel to the top of the semiconductor layer and is perpendicular to the fin length such that the gate conductor intersects a portion of the fin. An insulator (e.g., gate oxide) separates the gate conductor from the fin. Further, the region of the fin that is positioned below the gate conductor defines a semiconductor channel region. The FinFET structure can include multiple fins, in which case the gate conductor would wrap around, as well as fill in, the space between these fins.

(5) The fins extend across the active area of the semiconductor layer into where the raised source/drain regions are to be formed. A selective epitaxial growth/deposition process is used to form the raised source/drain regions. The raised source/drain regions typically comprise epitaxially grown silicon (Si) or silicon germanium (SiGe), for example.

(6) More particularly, epitaxially growing Si and SiGe facets may not form a rectangular profile on a silicon substrate having a (001) crystallographic orientation with a notch aligned in a $\langle 110 \rangle$ direction. Facets of the fin structure may exhibit a diamond shaped profile, which often occurs in conventional processing. This makes it difficult for source/drain extension formations since diamond shaped epitaxy is difficult to drive the dopants in the channel for a good overlap.

(7) One approach to form raised source/drain regions is disclosed in U.S. Pat. No. 8,310,013, which uses a damascene process to form the facets of the fin structure, i.e., the raised source/drain regions of the FinFET. The damascene process can be utilized to form unique and/or arbitrary profiles of the fin structure including the facets. The damascene process can utilize a capping layer that is patterned to define a desired facet profile. The capping layer can provide improved profile control. For example, the facets may be formed having a rectangular profile. Nonetheless, there is still a need for other approaches to form raised source/drain regions for a FinFET.

SUMMARY

(8) A method of making a semiconductor device includes forming a fin mask layer on a semiconductor layer, forming a dummy gate over the fin mask layer, and forming source and drain regions on opposite sides of the dummy gate. The dummy gate may be removed and the underlying

fin mask layer may then be used to define a plurality of fins in the semiconductor layer. A gate is formed over the plurality of fins.

(9) Forming the dummy gate may comprise forming the dummy gate so that portions of the fin mask layer extend outwardly therefrom. The method may further comprise forming a dummy gate mask layer over the dummy gate, and removing the portions of the fin mask layer that extend outwardly from the dummy gate using the dummy gate mask layer.

(10) Forming the source and drain regions may comprise forming raised source and drain regions. Removing the portions of the fin mask layer that extend outwardly from the dummy gate advantageously allows the raised source and drain regions to be more easily formed. With the fins removed from outside of the gate, the raised source and drain regions may be formed similar to bulk with comparable quality and control. This helps to enable strain techniques, such as silicon-germanium and silicon-carbon. Dopant drive-in (i.e., anneal) of the raised source and drain regions without the fins in place also allows for better source and drain extension overlap control.

(11) The method may further comprise removing upper portions of the semiconductor layer on opposite sides of the dummy gate to define source and drain recesses, with the source and drain regions being formed in the respective source and drain recesses.

(12) Forming the source and drain regions may comprise forming the source and drain regions of a different material than the semiconductor layer. Forming the dummy gate may comprise forming an oxide layer and a polysilicon layer thereover.

(13) The method may further comprise forming sidewall spacers on the dummy gate. The method may further comprise optionally removing the underlying fin mask layer before forming the gate.

(14) A semiconductor device may include a semiconductor layer, a plurality of semiconductor fins extending upwardly from the semiconductor layer and being spaced apart along the semiconductor layer, with each semiconductor fin having opposing first and second ends. A fin mask layer may be on the plurality of semiconductor fins. A gate may overlie the plurality of semiconductor fins and the fin mask layer, and have a width aligned with the opposing first and second ends of the plurality of semiconductor fins. Source and drain regions may be on opposite sides of the gate.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a flowchart illustrating a method for making a semiconductor device in accordance with the present embodiment.

(2) FIGS. 2 and 3 are top and cross-sectional side views showing formation of a fin mask layer on a semiconductor layer in accordance with the present embodiment.

(3) FIGS. 4, 5, and 6 are top and cross-sectional side views showing formation of a dummy gate over a portion of the fin mask layer fin shown in FIG. 2.

(4) FIGS. 7, 8, 9, and 10 are top and cross-sectional side views showing removal of the portions of the fin mask layer that extend outwardly from the dummy gate shown in FIG. 4.

(5) FIG. 11 is a cross-sectional side view showing sidewall spacers added to the dummy gate shown in FIG. 10, and the formation of source and drain recesses.

(6) FIG. 12 is a cross-sectional side view showing raised source and drain regions formed in the source and drain recesses shown in FIG. 11.

(7) FIGS. 13, 14, and 15 are top and cross-sectional side views showing dielectric deposition and chemical mechanical planarization (CMP) after forming the source and drain regions shown in FIG. 11.

(8) FIGS. 16, 17, and 18 are top and cross-sectional side views showing formation of the fins in the semiconductor layer using the fin mask layer shown in FIGS. 13, 14 and 15.

(9) FIG. 19 is a cross-sectional side view showing removal of the fin mask layer from the fin

shown in FIG. 17.

(10) FIGS. 20 and 21 are cross-sectional side views showing the gate formed over the fins shown in FIGS. 17 and 18.

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENTS

(11) The present embodiments will now be described more fully hereinafter with reference to the accompanying drawings, in which preferred embodiments are shown. The embodiments may, however, be embodied in many different forms and should not be construed as limited to the embodiments set forth herein. Rather, these embodiments are provided so that this disclosure will be thorough and complete, and will fully convey the scope of the disclosure to those skilled in the art. Like numbers refer to like elements throughout.

(12) A method of making a semiconductor device will now be discussed in reference to the flowchart 200 in FIG. 1 and to the process flow illustrated in FIGS. 2-11. As will be discussed in greater detail below, the fins are removed from outside of the gate so that the raised source and drain regions may be more easily formed. This helps to enable strain techniques, such as silicon-germanium and silicon-carbon. Dopant drive-in (i.e., anneal) of the raised source and drain regions without the fins in place also allows for better overlap control.

(13) Referring initially to the flowchart 200 in FIG. 1 and to the process flow illustrated in FIGS. 2 and 3, the method includes, from the start (Block 202), forming a fin mask layer 36 on a semiconductor layer 34 at Block 204. The illustrated process flow includes forming the fin mask layer 36 across an active area of the semiconductor layer 34.

(14) A top view of the fin mask layer 36 on the semiconductor layer 34 is illustrated in FIG. 2, and a cross-sectional side view along line AA' is illustrated in FIG. 3. The illustrated semiconductor layer 34 is on a dielectric layer 32 and is configured as a semiconductor on insulator (SOI). The dielectric layer 32 is on a semiconductor layer 30. As an example, the semiconductor layers 30 and 34 are silicon, the dielectric layer 32 is a silicon dioxide or silicon oxide, and the fin mask layer 36 is silicon nitride.

(15) A dummy gate 44 is formed over a portion of the fin mask layer 36 at Block 206 so that portions 37 of the fin mask layer extend outwardly therefrom, as illustrated in FIG. 4. The dummy gate 44 is polysilicon, for example, and surrounds the upper and side surfaces of the fin mask layer 36. A dielectric layer 42 is between the fin mask layer 36 and the dummy gate 44. A dummy gate mask layer 46 is then formed over the dummy gate 44 at Block 208. A top view of the dummy gate mask layer 46 and dummy gate 44 over a portion of the fin mask layer 36 is illustrated in FIG. 4.

(16) A cross-sectional side view along lines AA' through a center of the dummy gate 44 is illustrated in FIG. 5. A cross-sectional side view along lines BB' through the portions 37 of the fin mask layer 36 that extend outwardly from the dummy gate 44 is illustrated in FIG. 6. The dummy gate mask layer 46 is polysilicon, and the dielectric layer 42 is silicon dioxide, for example.

(17) The portions 37 of the fin mask layer 36 that extend outwardly from the dummy gate 40 are removed at Block 210 using the dummy gate mask layer 46, as illustrated in FIG. 7. A reactive ion etching (RIE) process clears the active area of the fin mask layer 36 so that silicon only is in the area where the source and drain regions are to be formed. A top view of this process is illustrated in FIG. 7.

(18) A cross-sectional side view along lines AA' through a center of the dummy gate 44 is illustrated in FIG. 7. A cross-sectional side view along lines BB' with the portions 37 of the fin mask layer 36 removed is illustrated in FIG. 8. Also, a cross-sectional side view along lines CC' through the dummy gate 44 is illustrated in FIG. 10.

(19) Sidewall spacers 50 are formed on the dummy gate 44 at Block 212, as illustrated in 11. The sidewall spacers 50 are silicon nitride, for example, and protect the dummy gate 44 during formation of the source and drain regions. Upper portions of the semiconductor layer 34 on opposite sides of the dummy gate 44 are removed at Block 214 to define source and drain recesses 52, 54. The recesses 52, 54 are optional, but they provide better strain in the channel since the

amount of material in front of the channel is increased.

(20) Source and drain regions **62, 64** are formed at Block **216** in the source and drain recesses **52, 54**. A selective epitaxial growth/deposition process is used to form the raised source and drain regions **62, 64**. The raised source/drain regions **62, 64** typically comprise epitaxially grown silicon or silicon germanium, for example. In the illustrated embodiment, the raised source/drain regions **62, 64** are epitaxially grown silicon germanium.

(21) Removing the portions **37** of the fin mask layer **36** that extend outwardly from the dummy gate **44** advantageously allows the raised source and drain regions **62, 64** to be more easily formed. The raised source and drain regions **62, 64** may now be formed similar to bulk with comparable quality and control. This helps to enable strain techniques, as readily appreciated by those skilled in the art. Dopant drive-in (i.e., anneal) of the raised source and drain regions **62, 64** without the fins in place also allows for better source and drain extension overlap control.

(22) After the raised source and drain regions **62, 64** have been formed, a dielectric layer **66** is deposited, as illustrated in FIG. **13**. A chemical mechanical planarization (CMP) is then performed so that an upper surface of the dielectric layer **66** is coplanar with an upper surface of the dummy gate mask layer **46**.

(23) A top view after deposition and CMP of the dielectric layer **66** is illustrated in FIG. **13**. A cross-sectional side view along lines AA' through a center of the dummy gate **44** is illustrated in FIG. **14**. Also, a cross-sectional side view along lines BB' through the dummy gate **44** is illustrated in FIG. **15**.

(24) The dummy gate mask layer **46** and the dummy gate **44** are removed at Block **218** and the underlying fin mask layer **36** is used to define a plurality of fins **70** in the semiconductor layer **34**, as illustrated in FIG. **17**. A top view of the fins **70** underlying the fin mask layer **36** is illustrated in FIG. **16**. A cross-sectional side view along lines AA' through a center of the fins **70** is illustrated in FIG. **17**. Also, a cross-sectional side view along lines CC' through the fins **70** is illustrated in FIG. **18**. The fin mask layer **36** may optionally be removed at Block **220**, as illustrated in FIG. **19**.

(25) A gate **80** is formed over the plurality of fins **70** at Block **222**, as best illustrated in FIG. **21**. The gate **80** includes a polysilicon layer **82** on a dielectric layer **84**. The method ends at Block **224**.

(26) After completion of the above described process flow, a resulting semiconductor device, as best illustrated in FIGS. **20** and **21**, includes a semiconductor layer **34**, a plurality of semiconductor fins **70** extending upwardly from the semiconductor layer and being spaced apart along the semiconductor layer, with each semiconductor fin having opposing first and second ends. A fin mask layer **36** is on the plurality of semiconductor fins **70**. The fin mask layer **36** includes a plurality of spaced apart fin mask sections, with each fin mask section on a respective semiconductor fin **70**. A gate **80** overlies the plurality of semiconductor fins **70** and the fin mask layer **36**, and has a width aligned with the opposing first and second ends of the plurality of semiconductor fins, as best illustrated by the top view in FIG. **16**. Source and drain regions **62, 64** are on opposite sides of the gate **80**.

(27) Many modifications and other embodiments will come to the mind of one skilled in the art having the benefit of the teachings presented in the foregoing descriptions and the associated drawings. Therefore, it is understood that the disclosure is not to be limited to the specific embodiments disclosed, and that modifications and embodiments are intended to be included within the scope of the appended claims.

Claims

1. A semiconductor device, comprising: a semiconductor layer having a first portion defining a fin having a first height, a second portion at a first end of the fin in a first lateral direction, and a third portion at a second end of the fin in the first lateral direction, each of the second portion and the third portion having a second height lower than the first height; a gate overlying the fin; a source

region comprising a first epitaxially grown semiconductor on the second portion of the semiconductor layer; a drain region comprising a second epitaxially grown semiconductor on the third portion of the semiconductor layer, the source region and the drain region being at opposite ends of the gate in the first lateral direction; and in cross-sectional view, a first sidewall spacer on one side of the gate and a second sidewall spacer on the opposite side of the gate, the first and second sidewall spacers directly contacting opposite sides of the gate on sides nearest to the gate and directly contacting the source region and the drain region, respectively, on sides farthest from the gate, wherein: the fin has a lower portion below the second height and an upper portion above the second height, wherein the upper portion extends in the first lateral direction between the source region and the drain region; the fin has a width in a second lateral direction perpendicular to the first lateral direction; and each of the second and third portions of the semiconductor layer has a width in the second lateral direction.

2. The semiconductor device of claim 1, wherein the semiconductor layer has an opening on each side of the lower portion of the fin in the second lateral direction.

3. The semiconductor device of claim 1, further comprising a fin mask layer on the fin with the gate overlying the fin and the fin mask layer.

4. The semiconductor device of claim 1, wherein there is no fin mask layer on the fin underlying the gate.

5. The semiconductor device of claim 1, wherein the gate includes a dielectric layer and a polysilicon layer on the dielectric layer.

6. The semiconductor device of claim 5, wherein the dielectric layer is in direct contact with the fin.

7. The semiconductor device of claim 1, further comprising: an other semiconductor layer; and a dielectric layer on the other semiconductor layer, the semiconductor layer being on the dielectric layer and spaced from the other semiconductor layer by the dielectric layer.

8. The semiconductor device of claim 1, further comprising a dielectric layer on the source region and the drain region, the dielectric layer directly contacting the sidewall spacers.

9. The semiconductor device of claim 8, wherein a top surface of the dielectric layer is coplanar with an upper surface of the gate.

10. The semiconductor device of claim 1, wherein the fin does not extend laterally into the source and drain regions in the first lateral direction.

11. The semiconductor device of claim 1, wherein the fin does not extend laterally beyond the footprint of the gate and sidewall spacers in the first lateral direction.

12. The semiconductor device of claim 1, wherein: the semiconductor layer comprises a first semiconductor material; and the first and second epitaxially grown semiconductor structures comprise a second semiconductor material different from the first semiconductor material.

13. The semiconductor device of claim 12, wherein: the first semiconductor material is Si; and the second semiconductor material is SiGe.

14. A semiconductor device, comprising: a fin; an epitaxy source region on a first semiconductor and adjacent to a first end of the fin in a first lateral direction; an epitaxy drain region on a second semiconductor and adjacent to a second end of the fin in the first lateral direction; in cross-sectional view, a first sidewall spacer between the first end of the fin and the epitaxy source region and a second sidewall spacer between the second end of the fin and the epitaxy drain region in the first lateral direction; and a gate overlaying the fin, wherein: the first and second sidewall spacers directly contact opposite sides of the gate on sides nearest the gate and directly contact the source region and the drain region, respectively, on sides farthest from the gate; the fin extends between the epitaxy source region and the epitaxy drain region in the first lateral direction; the fin has a width in a second lateral direction perpendicular to the first lateral direction; and each of the first and second semiconductors has a width in the second lateral direction.

15. A semiconductor device, comprising: a semiconductor layer including (i) a fin and (ii) a first

recess and a second recess on two opposite ends of the fin in a first lateral direction, the fin having a top surface that is different from a bottom surface of each of the first recess and the second recess of the semiconductor layer, the fin protruding higher than the bottom surface; a gate overlying the top surface of the fin; a source region of an epitaxy semiconductor material in the first recess of the semiconductor layer; a drain region of the epitaxy semiconductor material in the second recess of the semiconductor layer; and in cross-sectional view, a first sidewall spacer on one side of the gate and a second sidewall spacer on the opposite side of the gate, the first and second sidewall spacers directly contacting opposite sides of the gate on sides nearest to the gate and directly contacting the drain region and the source region, respectively, on sides farthest from the gate, wherein: the fin extends in the first lateral direction between the source region and the drain region; the fin has a width in a second lateral direction perpendicular to the first lateral direction; and each of the first and second recesses has a corresponding width in the second lateral direction.
